

Foundation of Networks

Required components of a (computer) network:

1. **At least 2 terminal devices (end points, nodes):**
 - There must be an intention from this device connection (exchange data, share resources)
 - A network service: is something useful that the network lets you do, like exchanging data (communication) or for sharing resources (a common printer, a common database that all the employees at a company use)
2. **Transmission media.**
 - Wire based: These use guided signals that travel through the wires
 - Electrical wires (copper): used for telephone networks and LANs, these are inexpensive but used for limited distance.
 - Fibre-Optical cables: transmits data as light signals, used for data centers and the internet infrastructure but more expensive than electrical wires.
 - Wireless: These use electro magnetical waves
 - Examples like the wifi, Bluetooth and satellite.
 - No physical wires needed, High mobility
 - Used in phones, IoT
3. **Network Protocols:** A network protocol is a set of rules and specifications that define how devices communicate with each other.

History of Computer Networks:

- ARPANET was the first decentralized computer network, meaning it had no single central control. If one part failed, the rest could still work.
- In 1969, it connected four universities in the United States, allowing researchers to share data and computing power.
- In 1971, it was renamed DARPANET (after DARPA).
- It used existing telephone lines, which were unreliable, so a new method called packet switching was developed.
 - Data was split into small packets. Each packet could take a different path to the destination.
- In **1970s**, terminals were connected to mainframe computers, enabling multi-user systems (many users using one computer at the same time).
- In **1974**, IBM introduced System Network Architecture (SNA), a proprietary networking architecture for connecting IBM systems.

Parallel Data Transmission:

Means the data is sent at the same time, multiple bits are being sent at the same time, so if you have 8 wires you will have then 8 bits travelling through the bus at the same time.

Parallel data transmission offers higher bandwidth. Since many bits travel at the same time. Over the short distance its very fast and effective.

Local Bus Systems: data transmission inside computers are used with short buses these buses are called local buses. Short distance.

In a computer, a bus is a group of wires that connects components: CPU, RAM.

Since we use a lot of wires for the parallel transfer its very expensive.

Control Lines: Sending multiple bits at the same time requires coordination, otherwise the receiving device won't know when to read the data or what to do with it.

Clock = the signal that tells the receiver "look now, the data is ready.", Without it the receiver might read the data too early or too late, resulting in errors. The sender puts the data on the bus in sync with the clock, and at each clock pulse, the receiver reads all the bits at the same time.

Read: The data is being read a device

Write: the data is being sent to a device

Examples: ISA, ATA, SCSI, IEEE 1284 (printer port) all of them are local buses for computer components, and sometimes outer devices.

N.B: the parallel port of 8 bits has 25 pins (holes) only 8 of those holes carry data from 2-9 the rest are just control pins that has control signals. In case of 18 bits you might need 36 or more holes.

Serial Data Transmission:

Serial data transmission sends data bit by bit using only one bus.

Serial is used both inside computer (local bus) and in computer networks.

Serial data transmission has an advantage that makes it better for long distance data transfer since it has no skew (skew means that the data arrives at different times).

The disadvantage is that it is slower than parallel since data is transmitted bit by bit. (less bandwidth or throughput)

Examples

- RS-232: classic serial port for computers.
- IEEE 488: used in lab instruments (GPIB).
- USB: serial bus for peripherals.
- CAN: automotive serial bus.
- Fibre Channel: fast serial bus for storage area networks.
- Ethernet: network communication, serial at the electrical level.

Pros and Cons of Parallel vs. Serial Data Transmission

Parallel Data Transmission

- **Pros:**
 - High data transfer rate over short distances (multiple bits sent simultaneously)
 - Simple to implement for internal computer buses
- **Cons:**
 - Expensive and bulky due to many wires
 - Cross talks and skew limit long-distance use

Serial Data Transmission

- **Pros:**
 - Can transmit over long distances with minimal interference
 - Cheaper and simpler wiring (fewer wires)
 - Modern serial interfaces can achieve very high speeds
- **Cons:**
 - Lower raw data transfer rate per wire compared to parallel (one bit at a time)
 - Slightly more complex timing or encoding may be needed for high-speed serial communication

Categories of Network Extensions:

Personal Area Network (PAN): A network designed to connect devices close to one person usually within a few meters from 1-10 meters. Examples: wifi, Bluetooth, USB for connecting mouses keyboards and cameras.

Local Area Network (LAN): A network that connects computers and devices within a small area, typically a building, office, or campus.

Purpose: Share resources like files, printers, and internet connections.

It covers area from 500-1000 meters.

Examples: WLAN Wireless local area network (default wifi), Ethernet: most common wired LAN Standard it uses optic cables. Token ring older LAN that allows only one device that is holding a token to send data and is replaced now by ethernet.

Metro politan area Network (MAN): A Metropolitan Area Network (MAN) connects multiple Local Area Networks (LANs) across a city or metropolitan area up to 100km.

Purpose: Allows organizations or service providers to connect offices, campuses, or public networks within a city.

Its preferred to use fiber optic wires cause it offers high speed and more reliable for valid data.

WiMAX uses towers or base stations to transmit wireless broadband signals across a city or metropolitan area.

Wide Area Network (WAN):A network that spans large geographic areas, such as cities, countries, or even continents.

Purpose: Connect multiple LANs and MANs over long distances, enabling communication and data transfer between geographically separated networks up to 1000km.

Asynchronous Transfer Mode (ATM):

A WAN technology that transfers data in small fixed-size packets called cells.

Designed for high-speed data, voice, and video communication over long distances.

Provides low latency and predictable performance. Through 10GB.

Global Area Network (GAN):

A Global Area Network (GAN) is a network that spans the entire world or multiple countries and continents.

Purpose: Connect LANs, MANs, and WANs globally, enabling worldwide communication.

Its nearly unlimited: it can access any network that any (LAN, MAN, WAN) can access.

It connects using satellites. Or Fiber-optic cables Laid across land and under the sea (submarine cables) That Provide extremely high-speed and high-capacity data transfer.

Directional Dependence of Data Transmission:

Simplex: The data transfers only in one direction so we have a transmitter that sends waves or signals and a receiver that receives these signals.

After the data has been transmitted another sender can use the channel.

An example: Radio.

Half Duplex: data transfers in both ways, think of it as 2 computers that are sending data back and forth to each other. One of them sends data and the other receives and wait until the first has finished so it can use the same channel to transfer its data. It has a controller for when which computer has to send.

Example: one channel network like a walkie talkie.

Some fiber optic wires.

Full Duplex: The information travels both ways, it has double the bandwidth of the Half duplex. It offers channel for sending and another one for receiving so the 2 terminal devices can send and receive data at the same time.

Examples: Phone, networks with twisted pair cables because they provide separate wires for send and receive

Network Topologies:

Network topology is the layout or arrangement of devices (computers, switches, routers, etc.) in a network and how they are connected. it also shows how data flows between them.

We have many topologies and some times we combine them (hybrid topology).

Physical Topology: is how the network looks and how the computers are actually connected to each other (how the look connected to each other using cables and layout)

Logical Topology: is how the data actually flows and the pattern in the network from computer A to computer C you might have a tree topology physically but logically the data travels in a Bus way.

Graphical Topology:

Nodes: Represent devices like computers, printers, switches, or routers.

Edges: Represent connections between devices, like cables or wireless links.

Peer-to-peer Connection:

A direct connection between only 2 terminal devices (end devices) without a need of a hub or a switch.

Uni and Bi directional communication: means that data can flow only in one way (uni-directional) a phone can send another phone a file, the other phone can just receive.

Bi-Directional: both phones can send files and receive at the same time, a peer to peer chat.

Advantages: no need for infrastructure, cheap, fast for direct communication.

Disadvantages: security issues can arise, devices needs to be always on and connected, powered.

Fully Connected (Mesh):

Its an extension of the idea peer to peer since each node is connected to each other node directly.

Each node has connection with all nodes except itself. So let N be nodes numbers.

Connections of Node A is $N - 1$. so if we have 5 nodes A has 4 connections.

Number of connections in the entire mesh is equal $(n \cdot (n-1)) / 2$. $N= 5$, connections = 10

Advantages of Mesh Topology:

1. High Redundancy and Reliability
 - Multiple paths exist between any two nodes. If one link fails, data can travel through another path.
2. Scalability for Small Networks
 - Easy to add a new node (though it requires adding links to all existing nodes).
3. No Central Point of Failure
 - Unlike star topology, there's no single device whose failure shuts down the network.

Disadvantages of Mesh Topology

1. High Cost

- Requires many cables, interfaces, and ports.
- Expensive to set up, especially for large networks.

2. Unused Links in Partial Use

- Many connections may remain idle, wasting resources if the network is lightly used.

Ring:

In a ring topology, each device (node) is connected to exactly two other nodes, forming a closed loop (ring).

Data travels from node to node along the ring until it reaches its destination using a token, when the computer has the token then it's their turn to transfer data.

All nodes need to be active, every node must be able to receive and forward data. If one node fails or is turned off, it can break the ring unless redundancy mechanisms exist.

Broken wire healed by two-way communication Modern ring networks often use dual-ring or two-way communication. If a link or node fails, data can travel the other direction to reach its destination, keeping the network running.

Advantages: equal access for all nodes, no data collision since we use the tokens to control it, predictable performance.

Disadvantages: if a single end point fails it disrupts the entire system.

Bus or Linear Structure:

A bus topology is a network layout where all nodes are connected to a single main communication line (the bus).

Data travels along the bus from one end to the other.

All the end points are connected to the same bus, when an end point sends data to another one all the other computers (end points) get a copy of this data.

Cut in the bus: if the bus is cut the network is then split into 2 segments and neither of them can communicate with each other nor internally.

If the bus experiences a short-circuit, all communication on the network can stop.

Advantages: simple and inexpensive, easy to expand by simply adding nodes among the bus

Disadvantages: if the main cable (bus) fails the entire network goes down. Only one device can transmit at a time; collisions can occur in large networks.

Connecting Busses, Repeater vs. Bridge:

Repeater: when connecting two buses sometimes the signals get too weak on the end of the first bus, this is when we bring a repeater. A repeater simply amplifies the signals (makes it stronger) and then send it to the second bus.

Bridge: a bridge is different it also connects 2 buses and it also amplifies the signal but with extra functionality. A bridge can choose to block a signal if its unnecessary to travel to the next bus using the MAC address.

Star:

all nodes are connected to one modem, server called star coupler. The star working is essential if the star fails the entire networks fail (single point of failure)

its very easy to spot which device is failed, since it will be the only device affected.

Since we have a deterministic direction, device start device, we can decide on latency.

Passive Central Node: like old hubs, it just forwards the signals it does zero control over what passes, its very cheap but there is a high chance of collision.

Active Central Node: can route data, filter traffic. Its more expensive but more efficient and safe for the network.

Hub, Switch, Router:

Hub: it just takes the electrical signals from one end point and forward it to all the end points connected to it, it can only amplify the signals.

Switch: A switch is a bit better it can filter based on MAC address, so it can know which device is the data intended for and only direct the frames to it.

Router: A router is the gateway between the network and other network it can read the ip address and direct packets to another network or receive packets that is intended for the network that its connected to.

Tree:

A tree topology is built by connecting multiple star networks together.

There is: A root (top-level) node, Intermediate nodes (branches), End devices (leaves)

Long(er) distance transmissions supported: since we break the network into multiple hierarchy this makes it sustainable to send signals for longer distances, since each layer has a star coupler that amplifies the signals.

Advantages: Supports Longer Distances, Isolated Leaf Failures.

Disadvantages:

If a branch or higher-level node fails, all devices under it become unreachable. Complex Setup and Maintenance. Costly, Dependent on Backbone

Cellular Network:

Cell = a small area where devices can connect, they all have to communicate through the base of this cell.

Base station = the “central point” in that area. All devices talk to it.

Wireless = no wires; devices connect using radio signals.

One sender at a time = only one device can send data at a time in the cell, otherwise signals collide.

Examples:

- GSM (mobile phones)
- Wi-Fi (IEEE 802.11)
- Bluetooth

Network Topologies Nowadays:

Ring: Rarely Used

Bus: Industrial & Automotive Area CAN, PowerLAN

Star: Ethernet from switch to leaf (in modern Ethernet networks, basically all cables connect to a central node)

Tree : USB, Among Switches and Hubs using Ethernet

Mesh: Special wireless network modes, Logical topology connecting routers, Rarely used inside computer systems connecting computing nodes (cores)